SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA13 - THEORY OF COMPUTATION

Name- K Sravan Kumar

Reg.Number- 192311210

1. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with a and end with a

Code-

```
#include <stdio.h>
#include <string.h>

int main() {
  char str[100];
  int i, state = 0;

  printf("Enter a string: ");
  scanf("%99s", str);

  for (i = 0; str[i] != '\0'; i++) {
    if (state == 0) {
      if (str[i] == 'a')
        state = 1;
```

```
        else
            state = 3;
    }
    else if (state == 1) {
        if (str[i] == 'a')
            state = 1;
        else
            state = 2;
    }
    else if (state == 2) {
        if (str[i] == 'a')
            state = 1;
        else
            state = 2;
    }
    else {
        state = 3;
    }
}

if (state == 1)
    printf("String Accepted\n");
else
    printf("String Rejected\n");

return 0;
```

```
}
```

Enter a string: aba

String Accepted

2 Write a C program to simulate a Non-Deterministic Finite Automata (NFA) for the given language representing strings that start with o and end with 1

Code-

```
#include <stdio.h>
#include <string.h>

int main() {
   char str[100];
   int i, state = 0;

   printf("Enter a binary string: ");
   scanf("%99s", str);

   for (i = 0; str[i] != '\0'; i++) {
      if (state == 0) {
         if (str[i] == '0')
            state = 1;
         else
```

```
            state = 3;
        }
        else if (state == 1) {
            if (str[i] == '0')
                state = 1;
            else
                state = 2;
        }
        else if (state == 2) {
            if (str[i] == '0')
                state = 1;
            else
                state = 2;
        }
        else {
            state = 3;
        }
    }

    if (state == 2)
        printf("String Accepted\n");
    else
        printf("String Rejected\n");

    return 0;
}
```

OUTPUT-

Enter a binary string: 001

String Accepted

3 .Write a C program to find ε -closure for all the states in a Non-Deterministic Finite Automata (NFA) with ε -moves

```
#include <stdio.h>

int main() {
    int n, e, i, j, a, b;
    int trans[20][20] = {0};
    int closure[20][20] = {0};
    int visited[20], stack[20], top;

    printf("Enter number of states: ");
    scanf("%d", &n);

    printf("Enter number of epsilon transitions: ");
    scanf("%d", &e);

    printf("Enter epsilon transitions (from to):\n");
    for (i = 0; i < e; i++) {
        scanf("%d %d", &a, &b);
        trans[a][b] = 1;
    }
```

```
for (i = 0; i < n; i++) {
    for (j = 0; j < n; j++)
        visited[j] = 0;

    top = -1;
    stack[++top] = i;
    visited[i] = 1;

    while (top != -1) {
        a = stack[top--];

        for (b = 0; b < n; b++) {
            if (trans[a][b] && !visited[b]) {
                visited[b] = 1;
                stack[++top] = b;
            }
        }
    }

    for (j = 0; j < n; j++)
        closure[i][j] = visited[j];
}

printf("\nEpsilon-closure of all states:\n");

for (i = 0; i < n; i++) {
```

```
        printf("E-closure(q%d) = { ", i);
        for (j = 0; j < n; j++) {
            if (closure[i][j])
                printf("q%d ", j);
        }
        printf("}\n");
    }

    return 0;
}
```

Input-

Enter number of states: 4

Enter number of epsilon transitions: 3

Enter epsilon transitions (from to):

0 1

1 2

2 3

Output-

Epsilon-closure of all states:

E-closure(q0) = { q0 q1 q2 q3 }

E-closure(q1) = { q1 q2 q3 }

E-closure(q2) = { q2 q3 }

E-closure(q3) = { q3 }